Harjinder Kaur, Lovely Professional University

Unit 13: Clustering methods using Weka Tool

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Objectives

After this unit, you will be able to

- Understand the concept of unsupervised learning.
- Learn the various clustering algorithms.
- Know the difference between clustering and classification.
- Implementation of K-Means and Hierarchal clustering using Weka.

Introduction

Clustering is the organization of data in classes. However, unlike classification, it is used to place data elements into related groups without advanced knowledge of the group definitions i.e. class labels are unknown and it is up to the clustering algorithm to discover acceptable classes. Clustering is also called unsupervised classification because the classification is not dictated by given class labels. There are many clustering approaches all based on the principle of maximizing the similarity between objects in the same class (intra-class similarity) and minimizing the similarity between objects of different classes (inter-class similarity). Clustering can also facilitate taxonomy formation, that is, the organization of observations into a hierarchy of classes that group similar events together.

Example: For a data set with two attributes: AGE and HEIGHT, the following rule represents most of the data assigned to cluster 10:

If AGE >= 25 and AGE <= 40 and HEIGHT >= 5.0ft and HEIGHT <= 5.5ft then CLUSTER = 10

13.1 Unsupervised Learning

Unsupervised Learning, as clearly defined through its name, uses no classification or labeled information as input to the algorithm. Unsupervised learning is another machine learning method that uses unlabeled input data to discover patterns. Unsupervised learning aims to extract structure and patterns from unstructured data. There is no need for monitoring when learning unsupervised. Instead, it searches the data for patterns on its own.

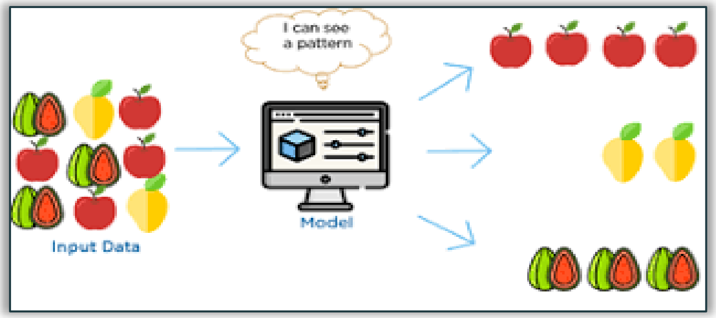


Figure 1: Unsupervised Learning

In unsupervised learning "The outcome or output for the given inputs is unknown"

Example: If the machine wants to differentiate between fishes and birds. If we do not tell the machine which animal is a fish and which is a bird (that is until we do not label them) machine will have to differentiate them by using similarities or patterns found in their attributes.

13.2 Introduction to Clustering

Clustering is the process of dividing a group of abstract items into classes of similar objects. Clustering is the grouping of similar objects, keeping in mind that: -

- objects of one cluster are similar to one another.
- objects of two different clusters differ from each other.

Clustering Requirements

The reasons why clustering is significant in data mining are as follows:

1. Scalability

To work with huge databases, we need highly scalable clustering techniques.

2. Ability to deal with a variety of attributes

Algorithms should be able to work with a variety of data types, including categorical, numerical, and binary data.

3. Cluster discovery with arbitrary shape

The method should be able to detect clusters of any shape and not be limited by distance measures.

4. Interpretability

The results should be complete, usable, and interpretable.

5. High dimensionality

Instead of merely dealing with low-dimensional data, the algorithm should be able to deal with high-dimensional space.

13.3 Clustering Methods

Clustering can be divided into two categories: hard clustering and soft clustering. One data point can only belong to one cluster in hard clustering. In soft clustering, however, the result is a probability likelihood of a data point belonging to each of the pre-defined clusters.

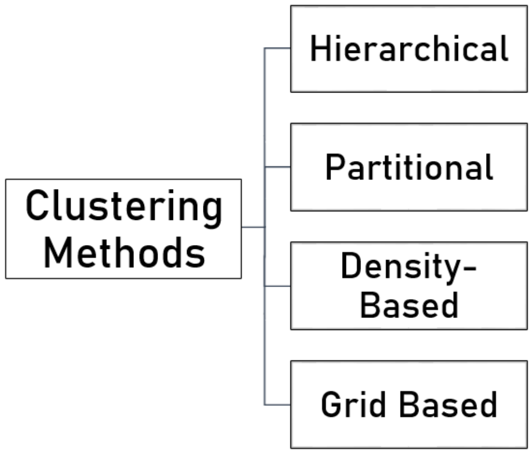


Figure 2: Clustering Methods

Hierarchical Clustering Methods

The hierarchical method decomposes the specified set of data items in a hierarchical manner. Methods can be classified based on how the hierarchical decomposition is generated. There are two approaches :

- Agglomerative
- Divisive

1. Agglomerative

The bottom-up technique is another name for this method. We begin by separating each object into its group. It continues to merge objects or groupings that are close together.

It continues to do so until all of the groups have been merged into one of the termination condition has been met.

2. Divisive

The opposite of Agglomerative, Divisive starts with all of the points in one cluster and splits them to make other clusters. These algorithms generate a distance matrix for all existing clusters and link them together based on the linkage criteria. A dendrogram is used to show the clustering of data points.

Partitional Method

This is one of the most popular methods for creating clusters among analysts. Clusters are partitioned based on the properties of the data points in partitioning clustering. For this clustering procedure, we must provide the number of clusters to be produced. These clustering algorithms use an iterative procedure to allocate data points between clusters based on their distance from one another. The following are the algorithms that fall within this category: –

1. K-Means Clustering

One of the most extensively used methods is K-Means clustering. Based on the distance metric used for clustering, it divides the data points into k clusters. The user is responsible for determining the value of 'k.' The distance between the data points and the cluster centroids is calculated. The cluster is assigned to the data point that is closest to the cluster's centroid. It computes the centroids of those clusters again after each iteration, and the procedure repeats until a pre-determined number of iterations have been completed or the centroids of the clusters have not changed after each iteration.

2. PAM (Partitioning Around Medoids)

The k-medoid algorithm is another name for this approach. It works similarly to the K-means clustering algorithm, except for how the cluster's center is assigned. The cluster's medoid must be an input data point in PAM, however, this is not true in K-means clustering because the average of all data points in a cluster may not be an input data point.

3. CLARA (Clustering Large Applications)

CLARA is a modification of the PAM method that reduces computing time to improve performance for huge data sets. To do so, it chooses a random portion of data from the entire data set to serve as a representation of the actual data. It uses the PAM algorithm to analyze several samples of data and selects the best clusters after several iterations.

Density-Based Methods

Clusters are produced using this method depending on the density of the data points represented in the data space. Clusters are locations that become dense as a result of the large number of data points that reside there.

The data points in the sparse region (the region with the few data points) are referred to as noise or outliers. These methods allow for the creation of clusters of any shape. Examples of density-based clustering methods are as follows:

1. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

The distance metric and criterion for a minimum number of data points are used by DBSCAN to group data points. It requires two inputs: eps and minimum points. The Eps value indicates how near data points should be to be considered neighbors. To consider the region as dense, the condition for minimum points should be completed.

Notes: The DBSCAN Clustering Algorithm's complexity

Best Case: We attain O(N log N) average runtime complexity if an indexing system is employed to store the dataset and neighborhood queries are done in logarithmic time.

Worst Case: Without using an index structure or working with degraded data (e.g., all points within a distance of less than), the worst-case run time complexity remains O(n2).

Average Case: Based on the data and algorithm implementation, the average case is the same as the best/worst case.

2. OPTICS (Ordering Points to Identify Clustering Structure)

It works similarly to DBSCAN, but it addresses one of the latter's flaws: the inability to generate clusters from data of arbitrary density. It also takes into account two other parameters: core distance and reachability distance. By selecting a minimal value for core distance, it is possible to determine if a data point is a core or not. The maximum core distance and the value of the distance metric used to calculate the distance between two data points are called reachability distances. One thing to keep in mind concerning reachability distance is that if one of the data points is a core point, its value is undefined.

Grid-Based methods

The collection of information is represented in a grid structure that consists of grids in grid-based clustering (also called cells). This method's algorithms take a different approach from the others in terms of their overall approach. They're more concerned about the value space that surrounds the data points than the data points themselves. One of the most significant benefits of these algorithms is their reduced computational complexity. As a result, it's well-suited to coping with massive data sets. It computes the density of the cells after partitioning the data sets into cells, which aids in cluster identification. The following are a few grid-based clustering algorithms: –

Statistical Information Grid Approach (STING)

The data set is partitioned recursively and hierarchically in STING. Each cell is subdivided further into a distinct number of cells. It records the statistical measurements of the cells, making it easier to respond to requests in a short amount of time.

CLIQUE (Clustering in Quest)

CLIQUE is a clustering technique that combines density-based and grid-based clustering. Using the Apriori principle, it partitions the data space and identifies the sub-spaces. It determines the clusters by calculating the cell densities.

13.4 Applications of Clustering

Many applications employ data clustering analysis. Market research, pattern recognition, data analysis, and picture processing are just a few examples.

Data clustering can also aid marketers in identifying separate client groupings. They can also categorize their customers based on their purchase habits.

It can be used to create plant and animal taxonomies in the realm of biology. Genes with comparable functions should be grouped to provide insight into population structures.

Clustering aids in the identification of areas in Data Mining. In an earth observation database, this is of comparable land usage. It can also be used to locate groupings of residences in a city. This varies depending on the type of home, its value, and its location.

Clustering in Data Mining also aids in the classification of web documents for information discovery.

Data clustering is also used in outlier identification applications. For example, detecting credit card fraud.

Cluster analysis is a data mining function that can be used as a tool. That is, to get knowledge about how data is distributed. It's also necessary to pay attention to the characteristics of each cluster.

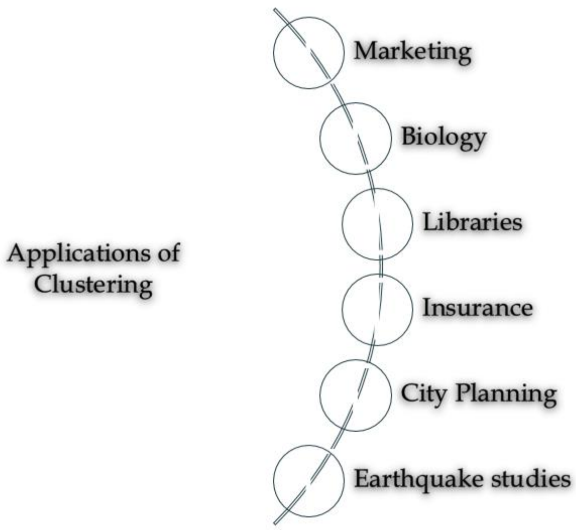


Figure 3: Application areas of clustering

13.5 The difference in Clustering and Classification

The classification of things into one or more classes based on features is done using both Classification and Clustering. They appear to be the same process since the differences are minor. In the case of classification, each input instance has predefined labels assigned to it based on its properties, whereas in the case of clustering, these labels are missing. The following table shows the difference between classification and clustering:

Table 1: Difference in classification and clustering

Parameter	Classification	Clustering
Type	Used for supervised learning	Used for unsupervised learning
Basic	Process of classifying the input instances based on their corresponding class labels	Grouping the instances based on their similarity without the help of class labels
Need	It has labels so there is a need for training and testing dataset	There is no need for training and testing dataset

	for verifying the model created	
Complexity	More complex as compared to clustering	Less complex as compared to classification
Example Algorithms	Logistic regression, Naive Bayes classifier, Support vector machines, etc.	k-means clustering algorithm, Fuzzy c-means clustering algorithm, Gaussian (EM) clustering algorithm, etc.

Differences between Classification and Clustering

1. Classification is used for supervised learning whereas clustering is used for unsupervised learning.

2. The process of classifying the input instances based on their corresponding class labels is known as classification whereas grouping the instances based on their similarity without the help of class labels is known as clustering.

3. As Classification have labels so there is need of training and testing dataset for verifying the model created but there is no need for training and testing dataset in clustering.

4. Classification is more complex as compared to clustering as there are many levels in the classification phase whereas only grouping is done in clustering.

5. Classification examples are Logistic regression, Naive Bayes classifier, Support vector machines, etc. Whereas clustering examples are k-means clustering algorithm, Fuzzy c-means clustering algorithm, Gaussian (EM) clustering algorithm, etc.

13.6 K-Mean Clustering Algorithm

The following steps show the working of the K-Means Clustering algorithm:

Step 1: Select a K value, where K denotes the number of clusters.

Step 2: Iterate through each point, assigning it to the cluster with the closest center. The centroid of all the clusters should be computed once each element has been iterated.

Step 3: Iterate through the dataset, calculating the Euclidean distance between each point and the cluster's centroid. If any point in the cluster is not nearest to it, reassign that point to the nearest cluster, and then calculate the centroid of each cluster again after doing so for all of the points in the dataset.

Step 4: Repeat Step 3 until there is no new assignment that occurred between the two iterations.

Task: What criteria can be used to decide the number of clusters in k-means statistical analysis?

Implementation of K-means in Weka

The following are the steps for implementing K-means using Weka:

1) Go to the Preprocess tab in WEKA Explorer and select Open File. Select the dataset "vote.arff."

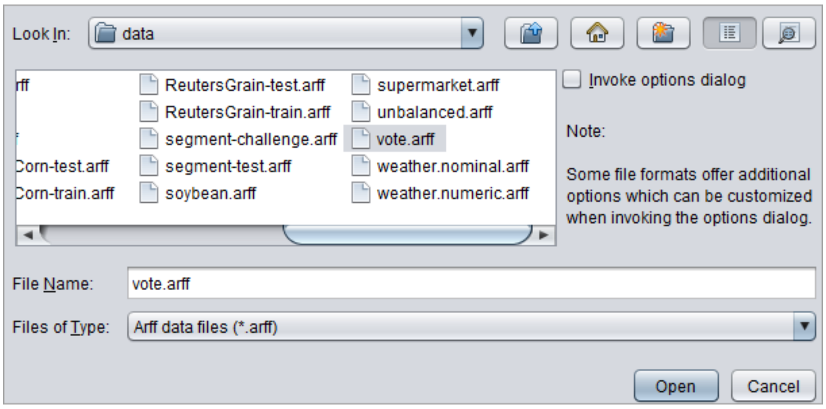


Figure 4: Dataset Selection

2) Select the "Choose" button from the "Cluster" menu. Choose "SimpleKMeans" as the clustering algorithm.

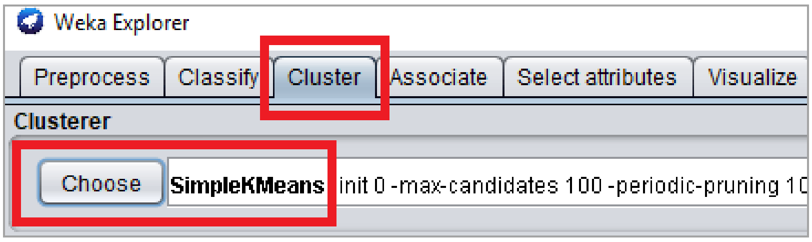


Figure 5: Selection of K-means

3) Select Settings, then fill in the following fields:

- Euclidian distance function
- There are six clusters in total. The sum of squared error will decrease as the number of clusters increases.

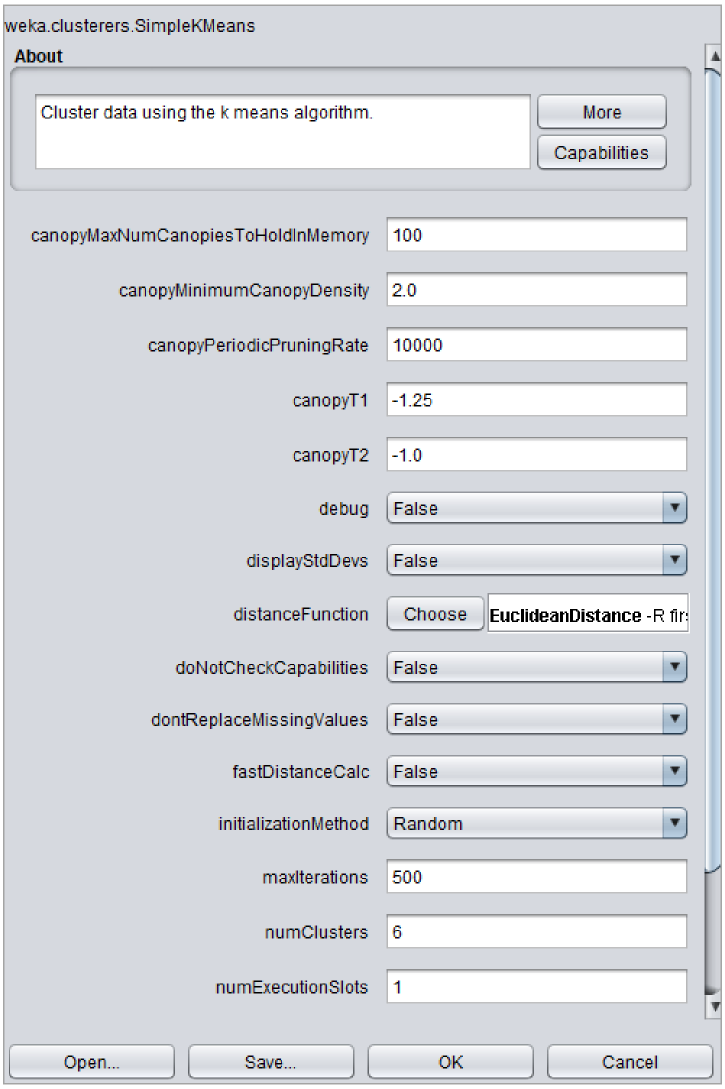
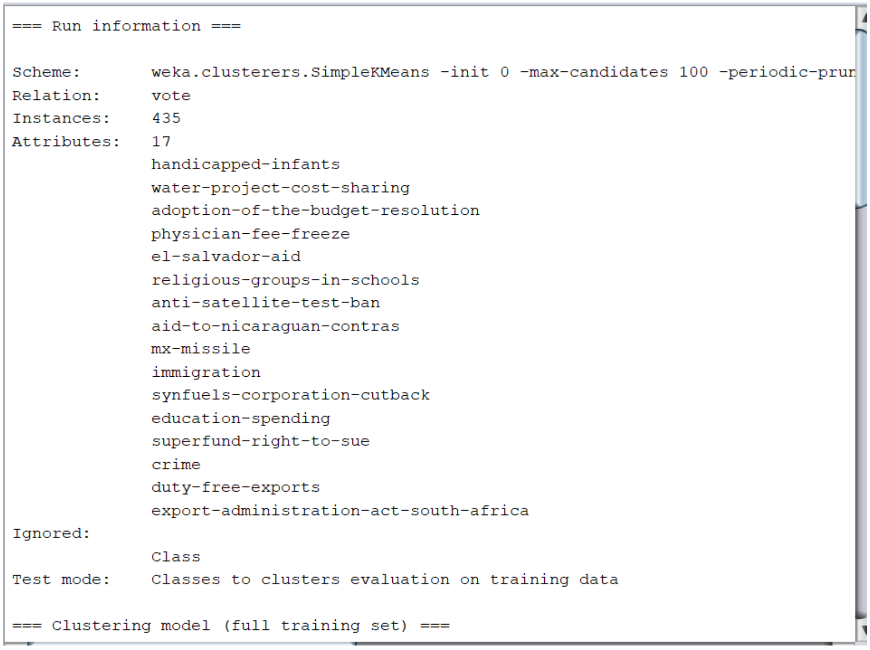


Figure 6: Parameters Setting

4. In the left panel, select Start. The algorithm's output is displayed on a white screen. Let us analyze the run information:

The properties of the dataset and the clustering process are described by the terms Scheme, Relation, Instances, and Attributes. The vote. arff dataset has 435 instances and 13 attributes in this scenario. The number of iterations with the Kmeans cluster is 5. The sum of the squared error is 1098.0. As the number of clusters grows, this inaccuracy will decrease. A table is used to represent the 5 final clusters with centroids. Cluster centroids are 168.0, 47.0, 37.0, 122.0.33.0, and 28.0 in our example.



```
=== Run information ===

Scheme:       weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-prun
Relation:     vote
Instances:    435
Attributes:   17
              handicapped-infants
              water-project-cost-sharing
              adoption-of-the-budget-resolution
              physician-fee-freeze
              el-salvador-aid
              religious-groups-in-schools
              anti-satellite-test-ban
              aid-to-nicaraguan-contras
              mx-missile
              immigration
              synfuels-corporation-cutback
              education-spending
              superfund-right-to-sue
              crime
              duty-free-exports
              export-administration-act-south-africa
Ignored:
              Class
Test mode:    Classes to clusters evaluation on training data

=== Clustering model (full training set) ===
```

Figure 7: K-means Run information

5. Choose "Classes to Clusters Evaluations" and click on Start.

The algorithm will assign the class label to the cluster. Cluster 0 represents republican and Cluster 1 represents democrat. The Incorrectly clustered instance is 14.023% which can be reduced by ignoring the unimportant attributes.

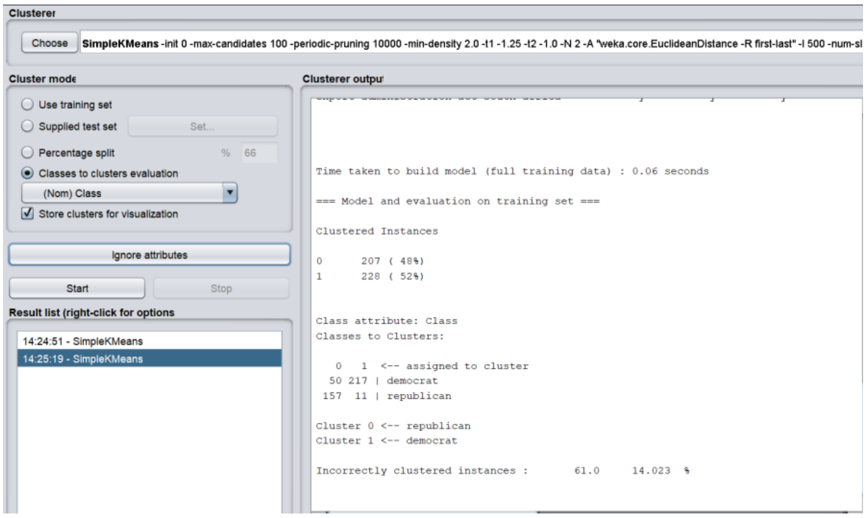


Figure 8: Cluster Formation using K-means

6. Use the "Visualize" tab to visualize the Clustering algorithm result. Go to the tab and click on any box. Move the Jitter to the max.

- The X-axis and Y-axis represent the attribute.
- The blue color represents class label democrat and the red color represents class label republican.
- Jitter is used to view Clusters.
- Click the box on the right-hand side of the window to change the x coordinate attribute and view clustering concerning other attributes.

Figure 9: Visualization of Result

> *Caution:* The number of clusters to use necessitates a precise balancing act. Larger k values can increase cluster homogeneity, but they also risk overfitting.

Implementation of Hierarchal clustering in Weka

1. Go to the Preprocess tab in WEKA Explorer and select Open File. Select the dataset "Iris.arff."

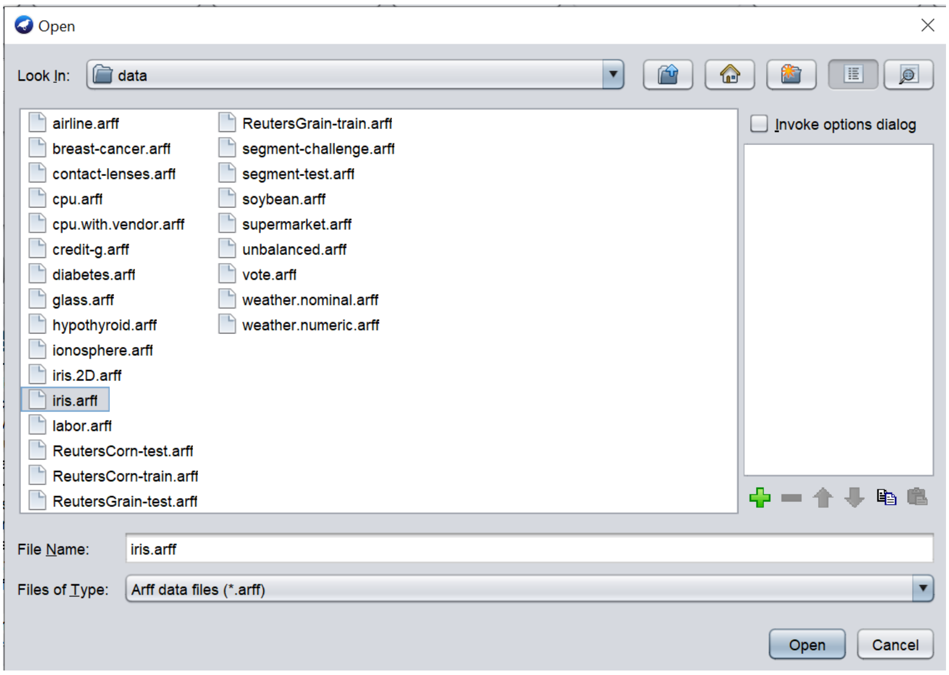


Figure 10: Selection of dataset

2. Select the "Choose" button from the "Cluster" menu. Choose "HierarichicalCluster" as the clustering algorithm.

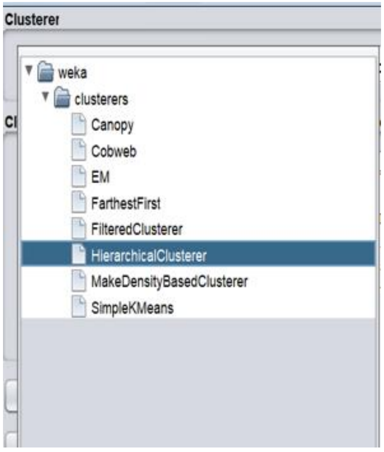


Figure 11: Selection of Algorithm

3. In the left panel, select Start. The algorithm's output is displayed on a white screen. Let us analyze the run information:

   The properties of the dataset and the clustering process are described by the terms Scheme, Relation, Instances, and Attributes. The Iris. arff dataset has 150 instances and 5 attributes in this scenario.

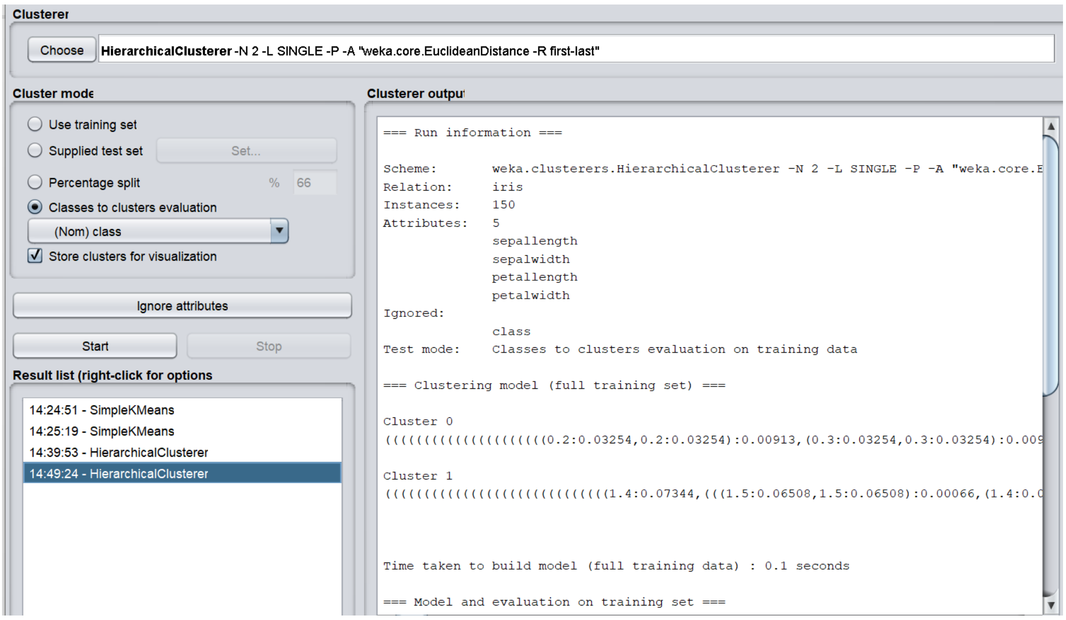


Figure 12: Run Information of Iris dataset

4. Choose "Classes to Clusters Evaluations" and click on Start.

The algorithm will assign the class label to the cluster. Cluster 0 represents Iris-setosa and Cluster 1 represents Iris-versicolor. The Incorrectly clustered instance is 33.3333 % which can be reduced by ignoring the unimportant attributes.

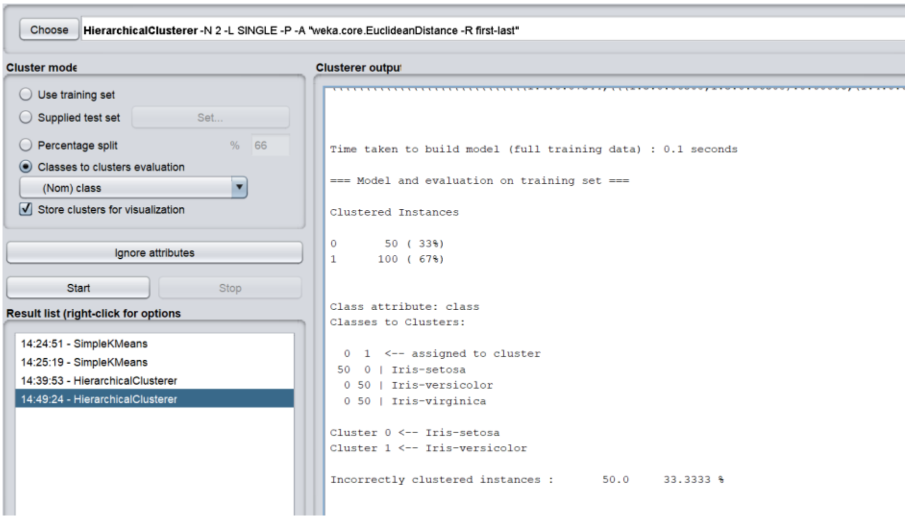


Figure 13: Cluster Formation

5 . Use the "Visualize" tab to visualize the Clustering algorithm result. Go to the tab and click on any box. Move the Jitter to the max. The X-axis and Y-axis represent the attribute.

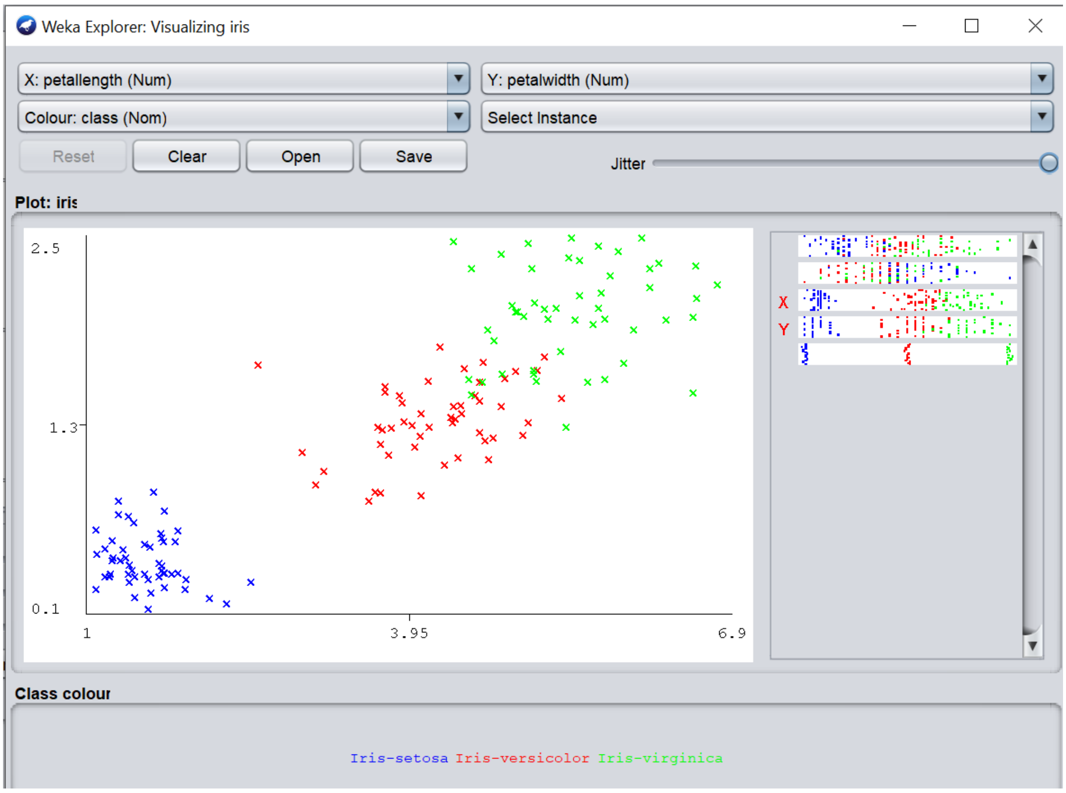


Figure 14: Visualization of Clusters

Task: Create your dataset and implement a hierarchical clustering algorithm on it and visualize the results.

Summary

- The learning of the model is 'supervised' if it is told to which class each training sample ⌘⌘belongs. In contrasts with unsupervised learning (or clustering), in which the class labels of the training samples are not known, and the number or set of classes to be learned may not be known in advance.

- The properties of the dataset and the clustering process are described by the terms Scheme, Relation, Instances, and Attributes.
- Classification is more complex as compared to clustering as there are many levels in the classification phase whereas only grouping is done in clustering.
- Data clustering can also aid marketers in identifying separate client groupings. They can also categorize their customers based on their purchase habits.
- One data point can only belong to one cluster in hard clustering. In soft clustering, however, the result is a probability likelihood of a data point belonging to each of the pre-defined clusters.
- Unsupervised learning is another machine learning method that uses unlabeled input data to discover patterns.

Keywords

Unsupervised Learning: Unsupervised Learning is a machine learning technique in which the users do not need to supervise the model.

Dendrogram: In a hierarchal clustering algorithm, we develop the hierarchy of clusters in the form of a tree, and this tree-shaped structure is known as the dendrogram.

Clustering: Grouping of unlabeled examples is called clustering.

Grid-based Methods: In this method, the data space is formulated into a finite number of cells that form a grid-like structure.

Similarity measure: You can measure similarity between examples by combining the examples' feature data into a metric, called a similarity measure.

High Dimensional Data: High-dimensional data is characterized by multiple dimensions. There can be thousands, if not millions, of dimensions.

Self Assessment Questions

1. _________________uses no classification or labelled information as input to the algorithm.

A. Unsupervised Learning

B. Supervised Learning

C. Reinforcement Learning

D. None

2. In unsupervised learning the outcome or output for the given inputs is

A. Known

B. Unknown

C. Available

D. Not Available

3. Unlabeled data is used in

A. Unsupervised Learning

B. Supervised Learning

C. Reinforcement Learning

D. None

4. Clustering is the grouping of similar objects, keeping in mind that

A. Objects of one cluster are similar to one another.

B. Objects of two different clusters differ from each other.

C. Both

D. None

5. DBSCAN is an example of which of the following clustering methods.

A. Density-Based Methods

B. Partitioning Methods

C. Grid-based Methods

D. Hierarchical Based Methods

6. STING is an example of which of the following clustering methods.

A. Density-Based Methods

B. Partitioning Methods

C. Grid-based Methods

D. Hierarchical Based Methods

7. Which of the following comes under applications of clustering.

A. Marketing

B. Libraries

C. City Planning

D. All of the above

8. In _____________ there is no need of training and testing dataset.

A. Classification

B. Clustering

C. Support vector Machines

D. None

9. Implementations of K-means only allow ____________values for attributes.

A. Numerical

B. Categorical

C. Polynomial

D. Text

10. The WEKA SimpleKMeans algorithm uses__________distance measure to compute distances between instances and clusters.

A. Euclidean

B. Manhattan

C. Correlation

D. Eisen

11. To perform clustering, select the _________tab in the Explorer and click on the _________button.

A. Choose, Cluster

B. Cluster, Choose

C. Create, Cluster

D. Cluster, Create

12. Click on the ___________tab to visualise the relationships between variables.

A. View

B. Show

C. Visualize

D. Graph

13. Examine the results in the Cluster output panel which give us:

A. The number of iterations of the K-Means algorithm to reach a local optimum.

B. The centroids of each cluster.

C. The evaluation of the clustering when compared to the known classes.

D. All of the Above

14. In ___________________ Weka can evaluate clustering on separate test data if the cluster representation is probabilistic.

A. In Supplied test set or Percentage split

B. Classes to clusters evaluation

C. Use training set

D. Visualize the cluster assignments

15. In this mode Weka first ignores the class attribute and generates the clustering.

A. In Supplied test set or Percentage split

B. Classes to clusters evaluation

C. Use training set

D. Visualize the cluster assignments

Review Questions

Q1) Briefly describe and give examples of each of the following approaches to clustering: partitioning methods, hierarchical methods, density-based and grid-based methods.

Q2) Elucidate the step-by-step working of K-Means with examples.

Q3) Explain the concept of unsupervised learning with examples. "Clustering is known as unsupervised learning" justify the statement with an appropriate example.

Q4) Differentiate between classification and clustering. Discuss the various applications of clustering.

Q5) With example discuss the various partitioning-based clustering methods.

Q6) Elucidate the various types of hierarchical clustering algorithms by giving the example of each type.

Answers: Self Assessment

1	A	2	B
3	A	4	C
5	A	6	C
7	D	8	B
9	A	10	A
11	B	12	C
13	D	14	A
15	B		

Further Readings

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